Derivatives: Contracts, Models, and Markets · Technical Note 30

Three Adjustments Done Exactly: the Timing Covariance, the Quanto Density and the Futures Convexity

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Chapter 30 of Derivatives: Contracts, Models, and Markets reduces every adjustment to one calculation: name the measure under which a variable's forward value is unbiased, name the measure the payoff requires, write the density between them, and take the covariance. This note takes three of the cases and carries that calculation out without the shortcuts the chapter allows itself. Part I removes the linearization from the timing adjustment: the exact result is a covariance with the discount factor *rather than with the rate, the dropped term is a co-skewness, and the exponential form the market quotes turns out to be exact rather than cosmetic when the discount factor is lognormal. It is also quantified—on the chapter's own example the first-order formula understates the adjustment by 6.5%, and the error runs one way. Part II carries out in full the change of numeraire that Section 30.6 states and checks by Itô's product rule. The density that change produces is proportional to the* reciprocal *of the exchange rate, which is where the minus sign in $\mu_V = r_f - \rho\sigma_V\sigma_S$ comes from; getting the inversion backwards is the commonest error in the derivation, and a one-sentence sign check catches it. Part III derives the chapter's two futures convexity formulas, Ho–Lee's (30.12) and Hull–White's (30.13), from a single integral of bond-price volatilities, and recovers the correction the chapter adds for the simply compounded rate.*

Part I. The timing adjustment without the linearization

1 THE EXACT STATEMENT

Work under Q^{T^*} and write

$$D \;=\; P(T^*,T), \qquad D_0 \;=\; \frac{P(0,T)}{P(0,T^*)} \;=\; \frac{1}{1+y_0\tau}$$

for the discount factor over the deferral period $[T^*,T]$ and its forward value today. Because $P(t,T)/P(t,T^*)$ is a traded asset over the numeraire, $E^{T^*}[D] = D_0$ exactly, and the density of Chapter 30's equation (30.4) evaluated at T^* is

$$\Lambda \;=\; \frac{D}{D_0} \;=\; \frac{1+y_0\tau}{1+y\tau}, \qquad E^{T^*}[\Lambda] = 1$$

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Therefore

$$E^T[V(T^*)] \;=\; E^{T^*}[V\Lambda] \;=\; \frac{E^{T^*}[VD]}{D_0} \;=\; V_0 \;+\; \frac{\mathrm{Cov}^{T^*}(V,D)}{D_0} \qquad \text{(TN30.1)}$$

with no approximation whatever. **The covariance is with the discount factor, not with the rate.** This is Chapter 30's equation (30.6); what follows is what the chapter does not do with it.

2 WHERE THE APPROXIMATION ENTERS

Chapter 30 replaces $\mathrm{Cov}(V,D)$ by a covariance with the rate. Writing $\Delta = y - y_0$ and expanding $\Lambda = 1/(1+u)$ with $u = \tau\Delta/(1+y_0\tau)$,

$$\Lambda \;=\; 1 \;-\; \frac{\tau\Delta}{1+y_0\tau} \;+\; \frac{\tau^2\Delta^2}{(1+y_0\tau)^2} \;-\; \cdots$$

so that

$$E^T[V] - V_0 \;=\; \underbrace{-\frac{\tau}{1+y_0\tau}\,\mathrm{Cov}^{T^*}(V,y)}_{\text{the term behind Chapter 30's (30.7)}} \;+\; \frac{\tau^2}{(1+y_0\tau)^2}\,\mathrm{Cov}^{T^*}\left(V,\Delta^2\right) \;-\; \cdots \qquad \text{(TN30.2)}$$

The dropped leading term is a *co-skewness*: the covariance of V with the *squared* rate move. It is second order in $\tau\sigma_y$, so in basis points the error grows with both the deferral and the rate volatility; relative to the adjustment, Section 4 shows, it is the volatility that matters.

3 WHEN THE EXPONENTIAL FORM IS EXACT

Equation (30.8) of Chapter 30 is usually presented as a cosmetic re-expression of (30.7). It is more than that. If V and D are jointly lognormal under Q^{T^*} with $E[V] = V_0$ and $E[D] = D_0$, then $E[VD] = V_0 D_0 \exp(\rho_{VD}\,\sigma_V\sigma_D T^*)$, and (TN30.1) gives

$$\boxed{E^T[V(T^*)] \;=\; V_0 \exp(\rho_{VD}\,\sigma_V\,\sigma_D\,T^*)} \qquad \text{(TN30.3)}$$

exactly, where σ_D is the volatility of the discount factor and ρ_{VD} its correlation with V. Chapter 30's exponent is this one with the first-order translations

$$\frac{dD}{D} = -\frac{\tau\,dy}{1+y\tau} \quad\Longrightarrow\quad \sigma_D \approx \frac{y_0\tau}{1+y_0\tau}\,\sigma_y, \qquad \rho_{VD} \approx -\rho$$

which reproduce $-\rho\,\sigma_V\sigma_y y_0\tau T^*/(1+y_0\tau)$ term for term. So the exponential form is exact in the discount-factor parametrization, and the approximation lives entirely in expressing σ_D and ρ_{VD} through the rate's volatility and correlation.

4 HOW MUCH IT COSTS

Take the parameters of Chapter 30's first example: $V_0 = 4.00\%$, $\sigma_V = 20\%$, $y_0 = 4.00\%$, $\sigma_y = 20\%$, $\rho = 0.7$, $T^* = 5$, $\tau = 0.25$.

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If the discount factor is lognormal. Then $\sigma_D = 0.0019802$, $\rho_{VD} = -0.7$, and (TN30.3) gives the exponent -0.00138614 and $E^T[V] = 3.9945\%$—identical to the chapter's answer to ten decimal places, because the chapter's formula *is* (TN30.3) under this assumption.

If the rate is lognormal. Then $D = 1/(1 + y\tau)$ is not lognormal, and (TN30.1) must be integrated. Fixing the rate's median so that $E[D] = D_0$ holds exactly:

	Exact	Chapter's (30.8)
$E^T[V]$	3.9941%	3.9945%
Adjustment	−0.5928 bp	−0.5541 bp

The approximation understates the adjustment by 6.5% of itself—about 0.04 basis points on this trade.

How the gap grows. Holding $T^* = 5$, $\rho = 0.7$ and $\sigma_V = 20\%$:

σ_y	τ	Exact (bp)	(30.8) (bp)	Error
20%	0.25	−0.5928	−0.5541	−6.5%
20%	1.00	−2.2780	−2.1481	−5.7%
20%	2.00	−4.3332	−4.1267	−4.8%
40%	0.25	−1.2571	−1.1074	−11.9%
40%	1.00	−4.6843	−4.2846	−8.5%
40%	2.00	−8.6655	−8.2109	−5.2%
60%	0.25	−1.9529	−1.6599	−15.0%
60%	1.00	−6.9125	−6.4096	−7.3%
60%	2.00	−12.3730	−12.2529	−1.0%

The error is dominated by the rate's volatility rather than by the deferral length: at $\sigma_y = 60\%$ and a quarterly deferral the first-order formula misses a sixth of the adjustment, while at 20% it is within a few percent whatever τ is. The sign never changes—the approximation always understates—which matters because, as Section 30.7 observes, an omitted adjustment biases every affected trade in one direction. So does an understated one.

Part II. The quanto adjustment, strictly by change of numeraire

5 WHAT HAS TO BE SHOWN

The setting has two economies, domestic and foreign. Write $B_d(t)$ and $B_f(t)$ for the two money market accounts, each started at 1 and accruing at its own short rate, and $S(t)$ for the exchange rate in *units of domestic currency per unit of foreign currency*. Let $V(t)$ be the foreign-currency price of a traded foreign asset paying no income, with

$$\frac{dV}{V} = \mu_V\,dt + \sigma_V\,dW_V, \qquad \frac{dS}{S} = \mu_S\,dt + \sigma_S\,dW_S, \qquad dW_V\,dW_S = \rho\,dt \quad \text{(TN30.4)}$$

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A quanto pays a function of $V(T)$ in domestic currency at a contractual exchange rate. Valuing it requires the law of V under the *domestic* risk-neutral measure Q^d, and what is known is its law under the *foreign* one, Q^f, where V is an ordinary traded asset. The claim to be established is

$$\mu_V = r_f - \rho\,\sigma_V\,\sigma_S \qquad \text{under } Q^d \tag{TN30.5}$$

6 THE TWO NUMERAIRES

A measure is named by its numeraire, so the first step is to say which asset each of the two measures divides by. Q^d is the measure associated with B_d: every domestic traded price divided by B_d is a Q^d-martingale.

Q^f needs one observation before it can be named in domestic terms. The foreign money market account is not a domestic asset, but *its domestic value is*: a domestic investor can buy foreign currency and roll it at the foreign short rate, and the position is worth

$$N(t) \;=\; S(t)\,B_f(t) \tag{TN30.6}$$

domestic units. N is therefore a traded domestic asset, strictly positive, and the measure associated with it is exactly Q^f: a foreign traded price X has domestic value XS, and

$$\frac{X(t)S(t)}{N(t)} = \frac{X(t)S(t)}{S(t)B_f(t)} = \frac{X(t)}{B_f(t)}$$

so the domestic-asset-over-N ratios are precisely the foreign-asset-over-B_f ratios that Q^f makes martingales. The foreign risk-neutral measure is the domestic measure associated with the numeraire SB_f. Nothing else about the foreign economy is needed.

7 THE DENSITY

The change-of-numeraire theorem states that for numeraires M and N, with Q^M and Q^N the measures they induce,

$$\left.\frac{dQ^M}{dQ^N}\right|_{\mathcal{F}_t} \;=\; \frac{M(t)\,/\,M(0)}{N(t)\,/\,N(0)} \tag{TN30.7}$$

Here the target measure is Q^d, so $M = B_d$, and the source is Q^f, so $N = SB_f$ by (TN30.6). With $B_d(0) = B_f(0) = 1$,

$$\boxed{\Lambda_t \;=\; \left.\frac{dQ^d}{dQ^f}\right|_{\mathcal{F}_t} \;=\; \frac{B_d(t)}{S(t)\,B_f(t)} \cdot S(0)} \tag{TN30.8}$$

Which way up this fraction goes is the whole derivation. Λ is proportional to $1/S$, not to S: moving *to* the domestic measure divides by the domestic value of the foreign account. The reciprocal is dQ^f/dQ^d, and using it produces (TN30.5) with the sign reversed. Section 9 gives the check that catches the error without redoing the algebra.

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8 GIRSANOV

Take the two short rates deterministic for the moment; Section 10 removes the assumption. Then B_d and B_f contribute no randomness and (TN30.8) reads

$$\Lambda_t = e^{(r_d - r_f)t}\,\frac{S(0)}{S(t)} \tag{TN30.9}$$

Under Q^f the exchange rate seen from the foreign side, $X = 1/S$ in units of foreign currency per unit of domestic, satisfies

$$\frac{dX}{X} = (r_f - r_d)\,dt + \sigma_S\,dW_X^f$$

by the same argument that gives S a drift of $r_d - r_f$ under Q^d: the domestic account is a foreign traded asset once converted. Substituting into (TN30.9),

$$\frac{d\Lambda}{\Lambda} = \sigma_S\,dW_X^f \tag{TN30.10}$$

and the drift has vanished, as it must—a Radon–Nikodym derivative is a martingale under the measure it starts from, and that cancellation is a check on (TN30.8) in its own right.

Girsanov's theorem now reads the drift change straight off the density's volatility. For any process Y, the Q^d drift of Y exceeds its Q^f drift by the instantaneous covariation of Y with Λ, per unit of each:

$$\mu_Y^d \;=\; \mu_Y^f \;+\; \frac{1}{Y\Lambda}\,\frac{d\langle Y, \Lambda\rangle}{dt} \tag{TN30.11}$$

Apply it to V. Under Q^f the foreign asset earns the foreign short rate, $\mu_V^f = r_f$. For the covariation, note that $X = 1/S$ moves against S, so dW_X^f and dW_V have correlation $-\rho$, and by (TN30.10)

$$\frac{1}{V\Lambda}\,\frac{d\langle V, \Lambda\rangle}{dt} = \sigma_V\,\sigma_S\,(-\rho) = -\rho\,\sigma_V\sigma_S$$

which with (TN30.11) gives (TN30.5):

$$\mu_V = r_f - \rho\,\sigma_V\,\sigma_S \qquad \text{under } Q^d$$

No product rule, no covered interest parity substitution, and no appeal to a hedging argument—only the numeraire, the theorem that turns it into a density, and Girsanov.

9 THE SIGN CHECK

The algebra can be trusted or not, but the sign can be settled in one sentence, and it is worth having because the inversion in (TN30.8) is the step that goes wrong.

Suppose $\rho > 0$: the foreign asset tends to be high exactly when foreign currency is expensive in domestic terms. A dealer short the quanto hedges with the foreign asset, whose gains arrive in foreign currency, while the payoff it owes is fixed in domestic units. With $\rho > 0$ those gains are largest exactly when foreign currency is worth most, so the hedge's domestic value outruns the payoff. That is a benefit to the hedger, it recurs, and competition passes it to the buyer as a lower forward—a drift *below* r_f,

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which is what (TN30.5) says and what $\Lambda \propto 1/S$ delivers. Had the density been taken proportional to S, the covariation would have been $+\rho\sigma_V\sigma_S$ and the drift would have exceeded r_f, charging the buyer for a benefit the hedge already provides.

10 STOCHASTIC RATES

The drift result does not in fact need deterministic rates. Taking logarithms in (TN30.8),

$$d\ln\Lambda = d\ln B_d - d\ln S - d\ln B_f$$

and since $d\ln B_d = r_d\,dt$ and $d\ln B_f = r_f\,dt$ carry no diffusion even when the short rates are stochastic, the density's diffusion is $-\sigma_S\,dW_S$ alone: (TN30.10) and (TN30.5) hold unchanged, with r_f the (now stochastic) foreign short rate. What stochastic rates change is the step from the drift to the forward in Section 11. A quanto forward settled at T is an expectation under the domestic T-forward measure, not under Q^d, and with stochastic rates the two differ; moving between them brings in the covariances of the asset with the domestic and foreign zero-coupon bonds, the same structure Part I meets as the timing adjustment. In practice these terms are second order against $\rho\sigma_V\sigma_S$ for equity quantos and are routinely dropped; for a quanto on a *rate*, they are not, and the full expression is what a cross-currency desk uses.

11 THE QUANTO FORWARD

Once the drift is known the forward follows in one line, which is why Chapter 30 states it immediately after. Under Q^d the asset grows at $\mu_V = r_f - \rho\sigma_V\sigma_S$, so

$$E^d[V(T)] = V(0)\,e^{(r_f-\rho\sigma_V\sigma_S)T}$$

while the ordinary foreign-currency forward is $F_0^f = V(0)e^{r_fT}$. Dividing,

$$F_0^{\text{quanto}} \;=\; F_0^f\,e^{-\rho\,\sigma_V\sigma_S\,T} \tag{TN30.12}$$

which is the chapter's result. A quanto option is then priced with Black's formula on F_0^{quanto}, discounted at the domestic rate—the measure change having done all of its work in the forward.

Part III. The futures convexity adjustment in Gaussian short-rate models

12 THE FUTURES RATE AS AN EXPECTATION

Section 30.3 of Chapter 30 shows that a contract settled daily is priced under the risk-neutral measure Q: the futures price is the Q-expectation of the settlement value. A contract quoted as 100 minus a rate is linear in that rate, so the futures rate is the Q-expectation of the rate it settles on. Write $\tau = T_2 - T_1$ and let $R(t)$ be the continuously compounded forward rate for $[T_1, T_2]$ seen at t,

$$\tau R(t) \;=\; \ln P(t,T_1) - \ln P(t,T_2) \tag{TN30.13}$$

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so that $R(0)$ is today's forward rate and $R(T_1)$ is the spot rate for the period. Then

$$\text{Futures} - \text{Forward} \;=\; E^{Q}[R(T_1)] - R(0)$$

and the adjustment is simply the expected drift of R under Q between today and T_1. No change of measure is needed; daily settlement has already put the futures rate under Q.

13 THE DRIFT OF THE FORWARD RATE

In a one-factor Gaussian short-rate model every zero-coupon bond follows, under Q,

$$\frac{dP(t,T)}{P(t,T)} \;=\; r\,dt \;-\; \sigma_P(t,T)\,dW \tag{TN30.14}$$

where the bond-price volatility $\sigma_P(t,T)$ is deterministic and vanishes at $t = T$. Itô's lemma gives $d\ln P(t,T) = \left(r - \tfrac{1}{2}\sigma_P(t,T)^2\right)dt - \sigma_P(t,T)\,dW$, and subtracting the T_2 bond from the T_1 bond in (TN30.13) cancels the short rate:

$$\tau\,dR \;=\; \frac{1}{2}\left[\sigma_P(t,T_2)^2 - \sigma_P(t,T_1)^2\right]dt \;+\; \left[\sigma_P(t,T_2) - \sigma_P(t,T_1)\right]dW$$

The dW term has zero expectation under Q, so

$$\boxed{\text{Futures} - \text{Forward} \;=\; \frac{1}{2\tau}\int_0^{T_1}\left[\sigma_P(t,T_2)^2 - \sigma_P(t,T_1)^2\right]dt} \tag{TN30.15}$$

The integrand is positive whenever the longer bond is the more volatile, which gives the sign Chapter 30 states: the futures rate lies above the forward rate. A model enters only through its bond volatilities, and the two formulas of the chapter are this integral for two choices of σ_P.

14 HO–LEE: EQUATION (30.12)

Under Ho–Lee, $dr = \theta(t)\,dt + \sigma\,dW$, a shock moves every rate by the same amount, and a bond with remaining life $T - t$ has $\sigma_P(t,T) = \sigma(T-t)$. Then

$$\sigma_P(t,T_2)^2 - \sigma_P(t,T_1)^2 = \sigma^2\left[(T_2-t)^2 - (T_1-t)^2\right] = \sigma^2\tau\,(T_1 + T_2 - 2t)$$

and

$$\int_0^{T_1}\sigma^2\tau\,(T_1+T_2-2t)\,dt = \sigma^2\tau\left[(T_1+T_2)T_1 - T_1^2\right] = \sigma^2\tau\,T_1T_2$$

so (TN30.15) gives

$$\text{Futures} - \text{Forward} \;=\; \frac{1}{2}\,\sigma^2\,T_1\,T_2$$

which is Chapter 30's equation (30.12). The product T_1T_2 comes from integrating a volatility gap that is itself proportional to the remaining life, which is why the adjustment grows with the square of maturity.

15 HULL–WHITE: EQUATION (30.13)

With mean reversion, $dr = (\theta(t) - ar)\,dt + \sigma\,dW$, the bond volatility is $\sigma_P(t,T) = \sigma B(t,T)$ with $B(t,T) = \left(1 - e^{-a(T-t)}\right)/a$. The longer bond's factor splits as

$$B(t,T_2) \;=\; B(t,T_1) \;+\; e^{-a(T_1-t)}\,B(T_1,T_2)$$

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so that

$$B(t,T_2)^2 - B(t,T_1)^2 = 2\,B(t,T_1)\,e^{-a(T_1-t)}\,B(T_1,T_2) \;+\; e^{-2a(T_1-t)}\,B(T_1,T_2)^2$$

$B(T_1,T_2)$ does not depend on t, and the two remaining integrals are elementary; with $u = T_1 - t$,

$$\int_0^{T_1} e^{-2a(T_1-t)}\,dt = \frac{1-e^{-2aT_1}}{2a}$$

$$\int_0^{T_1} B(t,T_1)\,e^{-a(T_1-t)}\,dt = \int_0^{T_1} \frac{e^{-au}-e^{-2au}}{a}\,du = \frac{\left(1-e^{-aT_1}\right)^2}{2a^2} = \frac{1}{2}\,B(0,T_1)^2$$

Hence

$$\int_0^{T_1} \left[B(t,T_2)^2 - B(t,T_1)^2\right] dt = \frac{B(T_1,T_2)}{2a}\left[B(T_1,T_2)\left(1-e^{-2aT_1}\right) + 2a\,B(0,T_1)^2\right]$$

and multiplying by $\sigma^2/(2\tau)$ as (TN30.15) requires,

$$\text{Futures} - \text{Forward} = \frac{1}{\tau}\,B(T_1,T_2)\left[B(T_1,T_2)\left(1-e^{-2aT_1}\right) + 2a\,B(0,T_1)^2\right]\frac{\sigma^2}{4a}$$

which is Chapter 30's equation (30.13). The limit $a \to 0$ that the chapter checks by hand needs no separate argument here: $\sigma B(t,T) \to \sigma(T-t)$, so the integrand of (TN30.15) tends to the Ho–Lee one. Mean reversion lowers the adjustment because $B(t,T)$ grows more slowly than $T-t$: long bonds are less volatile than Ho–Lee makes them, and the gap in (TN30.15) narrows.

16 THE SIMPLY COMPOUNDED RATE

Both results adjust the continuously compounded rate R. A contract settles on the simply compounded rate $L = \left(e^{\tau R}-1\right)/\tau$, and Itô's lemma gives

$$dL \;=\; e^{\tau R}\,dR \;+\; \frac{1}{2}\,\tau\,e^{\tau R}\,(dR)^2$$

To leading order ($e^{\tau R} \approx 1$) the futures–forward gap for L is the gap for R plus $\frac{1}{2}\tau\int_0^{T_1}(dR)^2/dt\;dt$. Under Ho–Lee the diffusion of R is $\left[\sigma_P(t,T_2) - \sigma_P(t,T_1)\right]/\tau = \sigma$, so the extra term is $\frac{1}{2}\sigma^2\tau T_1$, the amount Chapter 30 says to add for the quarterly rate. It is small because it grows with T_1 rather than with T_1T_2.

17 WHAT THE THREE PARTS HAVE IN COMMON

Part I starts from a density and asks what it costs to linearize it. Part II starts from a numeraire and asks which way up the density goes. Both are the same discipline: the adjustment is a covariance with an explicit object, and the object is worth writing down before it is approximated. Chapter 30's Itô check reaches (TN30.5) by imposing r_d as the drift of two domestic traded assets and solving—a verification, which confirms the answer without exhibiting what produces it. The density is what generalizes: name the numeraire a contract actually settles against, write (TN30.7), and the adjustment follows for contracts the chapter never mentions. Part III is the simplest case of all: daily settlement puts the futures rate under Q directly, so no density is needed, and the adjustment is the forward rate's own drift, fixed by the model's bond volatilities.

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